

For two stages

compute-bound
↓

1: prefill: easy to make compute-limited (good) by making

2: Generation: { Generation one token at a time ($CT=1$)! B/T large enough
 B is number of concurrent requests, hard to make large enough

logics for Arithmetic Intensity (AI) calculation steps.

- ①. compute total Flops
- ②. compute total bytes transferred
- ③. write down the ratio: $AI = \frac{\text{Flops}}{\text{bytes transferred}}$
- ④. interpret scaling behavior (limit reasoning)
- ⑤. result analysis: prefill: compute-limited (T large)
 decode: memory-limited ($T=1$)

high: ~~compute-limited~~
low: memory-limited

For Attention layers (only matrix multiplications considered in Flash Attention)

initial: Flops=0 bytes-transferred=0.

- query →
- steps 1: Read $Q(B \times T \times D)$, $K(B \times S \times D)$, $V(B \times S \times D)$ from HBM
bytes-transferred $= 2 \times B \times T \times D + 2 \times B \times S \times D + 2 \times B \times S \times D$
- Attention →
- 2: compute $A = Q(B \times T \times D) @ K(B \times S \times D)$
flops $= 2 \times B \times S \times T \times D$
- output →
- 3: compute $Y = \text{softmax}(A) (B \times S \times T \times K \times G) @ V(B \times S \times K \times H)$
flops $= 2 \times B \times S \times T \times D$
- 4: write $Y (B \times T \times H)$ to HBM
V: q-matmul, ..., k-matmul, ..., y-matmul

$$\text{bytes_transferred} = 2 * B * T * D$$

$$\Rightarrow \text{flops} = 4 * B * S * T * D$$

$$\text{byte_transferred} = 4 * B * S * T * D$$

$$\text{Arithmetic intensity} = (\text{flops} / \text{byte_transferred}). \text{simplify}()$$

$$\text{intensity} = S * T / (S + T)$$

$$1. \text{prefill: } T = S$$

$$\text{prefill_intensity} = \text{intensity.subs}(T, S). \text{simplify}()$$

$$\text{prefill_intensity} = S/2$$

$$2. \text{Generation: } T = 1$$

$$\text{generation_intensity} = \text{intensity.subs}(T, 1). \text{simplify}()$$

$$\text{generation_intensity} < 1 \quad \# \text{bad}$$

unlike MLPs, no dependence on B , so batching doesn't help

$\Rightarrow$ in MLP layers, every sequence hits the same MLP weights (W_{up} , W_{gate} , W_{down} don't depend on B).

$\Rightarrow$ in Attention layers, every sequence has its own vectors KV cache (Q, K, V all depend on B).

Summary:

* prefill is compute-bound, generation is memory-limited

* MLP intensity is B (requires concurrent requests),

Attention AI is 1 (impossible to improve)